Number 8 — Converting units of length, area and volume · Compound measures

Calculator allowed. Marks in brackets. ★ = harder question. Give answers to 3 s.f. where rounding is needed.

1 A photograph measures 15 cm by 10 cm. Find its area in m². [2]

> Convert the lengths first: $0.15 \times 0.10 = 0.015$ m². (Or 150 cm² $\div 10\,000 = 0.015$ m², since 1 m² $= 100^2$ cm².)

2 One litre is 1000 cm³. How many cubic millimetres are there in 2.5 litres? Give your answer in standard form. [3]

> 2.5 litres $= 2500$ cm³. 1 cm³ $= 10^3$ mm³, so $2500 \times 10^3 = 2.5 \times 10^6$ mm³.

3 Sound travels at 340 m/s in air. **a** How many kilometres does sound travel in one day? **b** A cube has sides of that length. Work out its volume in km³, in standard form to 3 s.f. [3]

> **a** One day $= 24 \times 60 \times 60 = 86\,400$ s; distance $= 340 \times 86\,400 = 29\,376\,000$ m $= 2.94 \times 10^4$ km. **b** $(2.9376 \times 10^4)^3 = 2.54 \times 10^{13}$ km³.

4 A person drinks about 730 litres of water a year. What is this in millilitres per minute? [2]

> 730 litres $= 730\,000$ ml. Minutes in a year: $365 \times 24 \times 60 = 525\,600$. Rate $= 730\,000 \div 525\,600 = 1.39$ ml per minute.

5 **a** A cyclist rides for 25 minutes at an average speed of 8 m/s. How far does she ride, in kilometres? **b** A sprinter runs 60 m in 7.5 s. What is his speed in km/h? [6]

> **a** 25 min $= 1500$ s; distance $= 8 \times 1500 = 12\,000$ m $= 12$ km. **b** $60 \div 7.5 = 8$ m/s. In one hour (3600 s) he would cover $8 \times 3600 = 28\,800$ m $= 28.8$ km, so 28.8 km/h.

6 **a** A brass ornament has density 8.5 g/cm³ and mass 1.7 kg. Find its volume. **b** A cubic metre of oak has a mass of 720 kg. Find the density of oak in g/cm³. [6]

> **a** Density $= \dfrac{\text{mass}}{\text{volume}}$, so volume $= \dfrac{\text{mass}}{\text{density}} = \dfrac{1700}{8.5} = 200$ cm³ (mass in grams to match g/cm³). **b** 720 kg $= 720\,000$ g; 1 m³ $= 100^3 = 10^6$ cm³. Density $= \dfrac{720\,000}{1\,000\,000} = 0.72$ g/cm³ (oak floats: less than water's 1 g/cm³).

7 Pressure $= \dfrac{\text{force}}{\text{area}}$. Copy and complete the table. [3]

force	area	pressure
60 N	2.5 m²	___ N/m²
___ N	4.8 m²	12.5 N/m²
1500 N	___ m²	300 N/m²

> Row 1: $60 \div 2.5 = 24$ N/m². Row 2: force = pressure × area $= 12.5 \times 4.8 = 60$ N. Row 3: area = force ÷ pressure $= 1500 \div 300 = 5$ m².

Total: 25 marks

Check: area units scale by the square of the length factor (1 m² $= 10^4$ cm²), volume units by the cube (1 m³ $= 10^6$ cm³ $= 10^9$ mm³) · m/s → km/h: multiply by 3.6 · match units before dividing (g with cm³, kg with m³) · density = mass ÷ volume, pressure = force ÷ area — rearrange with a formula triangle.